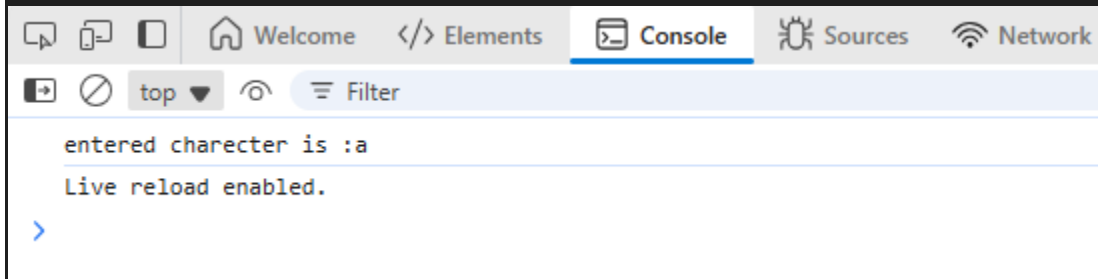


## Assignments

1. Accept a char input from the user and display it on the console.

```
let userInput = prompt('enter a character')[0];  
console.log('entered character is :'+userInput);
```



2. Accept two inputs from the user and output their sum.

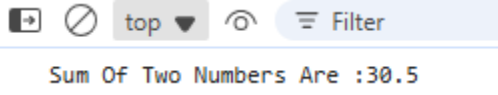
| Variable | Data Type |
|----------|-----------|
| Number 1 | Integer   |
| Number 2 | Float     |
| Sum      | Float     |

```
let firstNumber = parseInt(prompt("enter a number :"));
```

```

let secondNumber = parseFloat(prompt("enter a fractional
number :"));
let total = firstNumber + secondNumber;
console.log("Sum Of Two Numbers Are :"+total);ars

```



3. Write a program to find the simple interest.

- a. Program should accept 3 inputs from the user and calculate simple interest for the given inputs. Formula:  $SI = (P * R * n) / 100$

| Variable             | Data Type |
|----------------------|-----------|
| Principal amount (P) | Integer   |
| Interest rate (R)    | Float     |
| Number of years (n)  | Float     |
| Simple Interest (SI) | Float     |

```

let principleAmount = parseInt(prompt("enter your
principal amount :"));
let interestRate = parseFloat(prompt("enter your interest
rate :"));

```

```

let NumberOfYears = parseFloat(prompt("enter No of
years"));
let simpleInterest =
(principleAmonut*interestRate*NumberOfYears)/100;
console.log("Calculated Simple Interest
:"+simpleInterest);

```

Calculated Simple Interest :27000

4. Write a program to check whether a student has passed or failed in a subject after he or she enters their mark (pass mark for a subject is 50 out of 100).

- a. Program should accept an input from the user and output a message as “Passed” or “Failed”

| Variable | Data type |
|----------|-----------|
| mark     | float     |

```

let mark = parseFloat(prompt("Enter Your Mark :"));

if (mark >= 50 && mark <= 100) {
    console.log(mark+" Cangratulations you are passed ");
} else if (mark > 100) {
    console.log(mark+" is not a valid Score");
} else {
    console.log(mark+" Failed Try Again");
}

```

99 Cangratulations you are passed

5. Write a program to show the grade obtained by a student after he/she enters their total mark percentage.

- a. Program should accept an input from the user and display their grade as follows

| Mark  | Grade  |
|-------|--------|
| > 90  | A      |
| 80-89 | B      |
| 70-79 | C      |
| 60-69 | D      |
| 50-59 | E      |
| < 50  | Failed |

| Variable   | Data type |
|------------|-----------|
| Total mark | float     |

```

let totalMark = parseFloat(prompt("Enter your obtained score out
of 100"));

    if(totalMark >= 90 && totalMark <=100){
        console.log(totalMark+" Cangratulations you got A
grade");
    }else if(totalMark >=80 && totalMark <=89){
        console.log(totalMark+" Congartulations you got B
grade");
    }else if(totalMark >=70 && totalMark <=79){
        console.log(totalMark+" Congratulations you got C
grade");
    }else if(totalMark >=60 && totalMark <=69){
        console.log(totalMark+" Congratulation you got D
grade");
    }else if(totalMark >=50 && totalMark <=59){
        console.log(totalMark+" Congratulations you got E
grade");
    }else if(totalMark > 100){
        console.log("Enter a valid score obtained b/w 0 to
100");
    }else{
        console.log(totalMark+" You are failed try again");
    }

```

73 Congratulations you got C grade

6. Using the 'switch case' write a program to accept an input number from the user and output the day as follows.

| Input | Output |
|-------|--------|
| 1     | Sunday |

|                 |               |
|-----------------|---------------|
| 2               | Monday        |
| 3               | Tuesday       |
| 4               | Wednesday     |
| 5               | Thursday      |
| 6               | Friday        |
| 7               | Saturday      |
| Any other input | Invalid Entry |

```
let value = parseInt(prompt("enter a value to check corresponding
Day"));
switch(value){
  case 1:{
    console.log("Sunday");
    break;
  }
  case 2:{
    console.log("Monday");
    break;
  }
  case 3:{
    console.log("Tuesday");
    break;
  }
  case 4:{
    console.log("Wednesday");
    break;
  }
  case 5:{
    console.log("Thursday");
    break;
  }
}
```

```
    }  
    case 6:{  
        console.log("Friday");  
        break;  
    }  
    case 7:{  
        console.log("Saturday");  
        break;  
    }  
    default :{  
        console.log("Invalid entry");  
    }  
}
```

Monday

7. Write a program to print the multiplication table of given numbers.

a. Accept an input from the user and display its multiplication table

Eg:

**Output:** Enter a number

**Input:** 5

**Output:**

1 x 5 = 5

2 x 5 = 10

3 x 5 = 15

4 x 5 = 20

5 x 5 = 25

6 x 5 = 30

$$7 \times 5 = 35$$

$$8 \times 5 = 40$$

$$9 \times 5 = 45$$

$$10 \times 5 = 50$$

```
let number = parseInt(prompt("Enter a number for Multiplication
Table"));
for(i=1;i<=10;i++){
    console.log(i+" x "+number+" = "+i*number);
}
```

$$1 \times 5 = 5$$

$$2 \times 5 = 10$$

$$3 \times 5 = 15$$

$$4 \times 5 = 20$$

$$5 \times 5 = 25$$

$$6 \times 5 = 30$$

$$7 \times 5 = 35$$

$$8 \times 5 = 40$$

$$9 \times 5 = 45$$

$$10 \times 5 = 50$$

8. Write a program to find the sum of all the odd numbers for a given limit

- a. Program should accept an input as limit from the user and display the sum of all the odd numbers within that limit

For example if the input limit is 10 then the result is  $1+3+5+7+9 = 25$

**Output:** Enter a limit

**Input:** 10

**Output:** Sum of odd numbers = 25



```
let limit = parseInt(prompt("enter your limit"));
let sumOfOdd = 0;
for(let i=1;i<=limit;i+=2){
    sumOfOdd += i;
}
console.log("sum of odd number :"+sumOfOdd);
```

sum of odd number :25

9. Write a program to print the following pattern (**hint**: use nested loop)

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
let pattern = "";
for(let i=1;i<=5;i++){
    for(let j=1;j<=i;j++){
        pattern += j+" ";
    }
    pattern += "\n\n";
}
console.log(pattern);
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

10. Write a program to interchange the values of two arrays.

- a. Program should accept an array from the user, swap the values of two arrays and display it on the console

Eg: **Output:** Enter the size of arrays

**Input:** 5

**Output:** Enter the values of Array 1

**Input:** 10, 20, 30, 40, 50

**Output:** Enter the values of Array 2

**Input:** 15, 25, 35, 45, 55

**Output:** Arrays after swapping:

Array1: 15, 25, 35, 45, 55

Array2: 10, 20, 30, 40, 50

```
let array1 = [];  
    let array2 = [];  
    let limit = parseInt(prompt("enter the limit of the array  
:"));  
  
    for(let i=0;i<limit;i++){  
        let value = parseInt(prompt("enter the first array and  
"+ parseInt(i+1) +" position value "));  
        array1.push(value);  
    }  
    for(let j=0;j<limit;j++){  
        let value2 = parseInt(prompt("enter the second array  
and "+ parseInt(j+1) +" position value"));  
        array2.push(value2);  
    }
```

```
console.log("before swapping two arrays :");
console.log("first array "+array1);
console.log("second array "+array2);
for(let k=0;k<limit;k++){
    let temp = array1[k];
    array1[k] = array2[k];
    array2[k] = temp;
}
console.log("after swapping two arrays :")
console.log("first array "+array1);
console.log("second array "+array2);
```

```
before swapping two arrays :
first array 5,10,15,20,25
second array 30,35,40,45,50
after swapping two arrays :
first array 30,35,40,45,50
second array 5,10,15,20,25
```

11. Write a program to find the number of even numbers in an array

- a. Program should accept an array and display the number of even numbers contained in that array

Eg: **Output:** Enter the size of an array

**Input:** 5

**Output:** Enter the values of array

**Input:** 11, 20, 34, 50, 33

**Output:** Number of even numbers in the given array is 3

```
let array = [];
    let count = 0;
    let limit = parseInt(prompt("enter the limit of the array
:"));
```

```

        for(let i=0;i<limit;i++){
            let value = parseInt(prompt("enter the values of the
array :"));
            array.push(value);
        }
        for(let i=0;i<limit;i++){
            if(array[i]%2===0){
                count++;
            }
        }
        console.log("Total count of the even numbers in an array :"+
count);

```

Total count of the even numbers in an array :5

12. Write a program to sort an array in descending order

- a. Program should accept an array, sort the array values in descending order and display it

Eg: **Output:** Enter the size of an array

**Input:** 5

**Output:** Enter the values of array

**Input:** 20, 10, 50, 30, 40

**Output:** Sorted array:

50, 40, 30, 20, 10

```

let array = [];
let size = parseInt(prompt("enter the limit of the array
:"));
for(let i=0;i<size;i++){
    let value = parseInt(prompt("enter the values of the
array"));
    array.push(value);
}

```

```

    }
    for(let i=0;i<size-1;i++){
        for(let j=i+1;j<size;j++){
            if(array[i]<array[j]){
                let temp = array[i];
                array[i] = array[j];
                array[j] = temp;
            }
        }
    }
    console.log("Sorted array :"+array);

```

Sorted array :50,40,30,20,10

13. Write a program to identify whether a string is a palindrome or not

- a. A string is a palindrome if it reads the same backward or forward eg:

MALAYALAM

Program should accept a string and display whether the string is a palindrome or not

Eg: **Output:** Enter a string

**Input:** MALAYALAM

**Output:** Entered string is a palindrome

Eg 2: **Output:** Enter a string

**Input:** HELLO

**Output:** Entered string is not a palindrome

```

let name = prompt("enter a string");
let flag=0;
let length = name.length;
for(let i=0;i<length/2;i++){
    if(name[i] != name[length-i-1]){

```

```
        flag = 1;
        break;
    }
}
if(flag == 0){
    console.log(name+" is palindrome");
}else{
    console.log(name+" is not a palindrome ");
}
```

malayalam is palindrome

14. Write a program to add to two dimensional arrays

a. Program should accept two 2D arrays and display its sum

Eg: **Output:** Enter the size of arrays

**Input:** 3

**Output:** Enter the values of array 1

**Input:**

1 2 3

4 5 6

7 8 9

**Output:** Enter the values of array 2

**Input:**

10 20 30

40 50 60

70 80 90

**Output:** Sum of 2 arrays is:

11 22 33

44 55 66

77 88 99

```
let size = parseInt(prompt("enter the size of the array"));
let array1 = [];
for(let i=0;i<size;i++){
    array1[i] = [];
    for(let j=0;j<size;j++){
        array1[i][j] = parseInt(prompt("enter the values of
the array"));
    }
}

let array2 = [];
for(let i=0;i<size;i++){
    array2[i] = [];
    for(let j=0;j<size;j++){
        array2[i][j] = parseInt(prompt("enter the values of
the second array :"));
    }
}

console.log("before addition of arrays :")
console.log(array1);
console.log(array2);
let array3 = [];
for(let i=0;i<size;i++){
    array3[i] = [];
    for(let j=0;j<size;j++){
        array3[i][j] = array1[i][j] + array2[i][j];
    }
}

console.log("after addition of two arrays");
console.log(array3);
```

before addition of arrays :

```
▶ (3) [Array(3), Array(3), Array(3)] ⓘ  
  ▶ 0: (3) [1, 2, 3]  
  ▶ 1: (3) [4, 5, 6]  
  ▶ 2: (3) [7, 8, 9]  
    length: 3  
  ▶ [[Prototype]]: Array(0)
```

```
▶ (3) [Array(3), Array(3), Array(3)] ⓘ  
  ▶ 0: (3) [10, 20, 30]  
  ▶ 1: (3) [40, 50, 60]  
  ▶ 2: (3) [70, 80, 90]  
    length: 3  
  ▶ [[Prototype]]: Array(0)
```

after addition of two arrays

```
▶ (3) [Array(3), Array(3), Array(3)] ⓘ  
  ▶ 0: (3) [11, 22, 33]  
  ▶ 1: (3) [44, 55, 66]  
  ▶ 2: (3) [77, 88, 99]  
    length: 3  
  ▶ [[Prototype]]: Array(0)
```

15. Write a program to accept an array and display it on the console using functions

a. Program should contain 3 functions including main() function

#### main()

1. Declare an array
2. Call function `getArray()`
3. Call function `displayArray()`

#### getArray()

1. Get values to the array

#### displayArray()

1. Display the array values

```
function main() {
```



```

        let arr = [];
        let size = parseInt(prompt("enter the size of the
array :"));
        getArray(arr,size)
        displayArray(arr)
    }
    function getArray(array,len){
        for(let i=0;i<len;i++){
            array[i] = parseInt(prompt("enter the values of
the array :"));
        }
    }
    function displayArray(array){
        console.log("values of the array :"+array);
    }
    main();

```

values of the array :10,20,30,40,50

16. Write a program to check whether a given number is prime or not

- a. Program should accept an input from the user and display whether the number is prime or not

Eg: **Output:** Enter a number

**Input:** 7

**Output:** Entered number is a Prime number

```

let number = parseInt(prompt("enter a number to check prime or not
"));
let flag = 0;
for(let i=2;i<=number/2;i++){
    if(number%i==0){
        flag = 1;
        break;
    }
}

```

```

    }
    if(flag==0){
        console.log("given number is a prime number");
    }else{
        console.log("given number is not a prime number");
    }
}

```

given number is a prime number

17. Write a menu driven program to do the basic mathematical operations such as addition, subtraction, multiplication and division (**hint**: use if else ladder or switch)

- Program should have 4 functions named addition(), subtraction(), multiplication() and division()
- Should create a class object and call the appropriate function as user prefers in the main function

```

class product{
    constructor(value1,value2,choice){
        this.value1 = value1;
        this.value2 = value2;
        this.choice = choice;
    }
    addition(){
        console.log(`Answer :${this.value1+this.value2}`);
    }
    subtraction(){
        console.log(`Answer :${this.value1-this.value2}`);
    }
    multiplication(){
        console.log(`Answer :${this.value1*this.value2}`);
    }
    division(){
        console.log(`Answer :${this.value1/this.value2}`);
    }
}

```

```

    }

    function main(){
        let value1 = parseInt(prompt("enter a number :"));
        let value2 = parseInt(prompt("enter a another number
:"));
        let choice =
parseInt(prompt("1.addition\n2.substraction\n3.multiplication\n
4.division"));
        const objproduct = new product(value1,value2,choice);
        if(choice === 1){
            objproduct.addition();
        }else if(choice === 2){
            objproduct.substraction();
        }else if(choice === 3){
            objproduct.multiplication();
        }else if(choice === 4){
            objproduct.division();
        }else{
            console.log("please Select a valid option");
        }
    }
    main();

```

Answer :300

18. Grades are computed using a weighted average. Suppose that the written test counts 70%, lab exams 20% and assignments 10%.

If Arun has a score of

Written test = 81

Lab exams = 68

Assignments = 92

Arun's overall grade =  $(81 \times 70)/100 + (68 \times 20)/100 + (92 \times 10)/100 = 79.5$

Write a program to find the grade of a student during his academic year.

- Program should accept the scores for written test, lab exams and assignments
- Output the grade of a student (using weighted average)

Eg:

Enter the marks scored by the students

Written test = 55

Lab exams = 73

Assignments = 87

Grade of the student is 61.8

```
let writtenTest = 55;
    let labExam = 73;
    let assignment = 87;
    let gradeOf = parseFloat((writtenTest*70)/100 +
(labExam*20)/100 + (assignment*10)/100);
    console.log(`Written test = ${writtenTest}`);
    console.log(`Lab exams = ${labExam}`);
    console.log(`Assignments = ${assignment}`);
    console.log(`Grade of the student is ${gradeOf}`);
```

Written test = 55

Lab exams = 73

Assignments = 87

Grade of the student is 61.8

19. Income tax is calculated as per the following table

| Annual Income | Tax percentage |
|---------------|----------------|
|---------------|----------------|

|                            |        |
|----------------------------|--------|
| Up to 2.5 Lakhs            | No Tax |
| Above 2.5 Lakhs to 5 Lakhs | 5%     |
| Above 5 Lakhs to 10 Lakhs  | 20%    |
| Above 10 Lakhs to 50 Lakhs | 30%    |

Write a program to find out the income tax amount of a person.

- Program should accept annual income of a person

Output the amount of tax he has to pay

Eg 1:

Enter the annual income

495000

Income tax amount = 24750.00

Eg 2:

Enter the annual income

500000

Income tax amount = 25000.00

```
let anualIncome = parseFloat(prompt("Enter your Annual income"));

    if(anualIncome >= 250000 && anualIncome <= 500000){
        let tax = (5/100)*anualIncome;
        console.log(`Income tax calculated as per your annual
income ${tax.toFixed(2)}`);
    }
    else if(anualIncome > 500000 && anualIncome <= 1000000){
        let tax = (20/100)*anualIncome;
        console.log(`Income tax calculated as per your annual
income ${tax.toFixed(2)}`);
    }
```

```

    }
    else if(anualIncome > 1000000 && anualIncome <= 5000000){
        let tax = (30/100)*anualIncome;
        console.log(`Income tax calculated as per your annual
income ${tax.toFixed(2)}`);
    }
    else if(anualIncome < 2,50000){
        console.log(`you don't have to pay income tax`);
    }
    else{
        console.log('please enter amount below 5,000000');
    }

```

Income tax calculated as per your annual income 24750.00

20. Write a program to print the following pattern using for loop

```

1

2   3

4   5   6

7   8   9   10

```

```

let n = 4;
let numberPattern = "";
let count = 0;
for(let i=1;i<=n;i++){
    for(let j=1;j<=i;j++){
        numberPattern += ++count + " ";
    }
    numberPattern += "\n\n";
}
console.log(numberPattern);

```

1  
2 3  
4 5 6  
7 8 9 10

21. Write a program to multiply the adjacent values of an array and store it in an another array

- Program should accept an array
- Multiply the adjacent values
- Store the result into another array

Eg:

Enter the array limit

5

Enter the values of array

1      2      3      4      5

Output

2      6      12      20

```
let arr = [1,2,3,4,5];  
let arr2 = [];  
for(let i = 0;i<arr.length-1;i++){  
    arr2[i] = arr[i]*arr[i+1];  
}  
console.log(arr);  
console.log(arr2);
```

► (5) [1, 2, 3, 4, 5]

► (4) [2, 6, 12, 20]

22. Write a program to add the values of two 2D arrays

a. Program should contains 3 functions including the main function

**main()**

1. Call function `getArray()`
2. Call function `addArray()`
3. Call function `displayArray()`

**getArray()**

1. Get values to the array

**getArray()**

1. Add array 1 and array 2

**displayArray()**

1. Display the array values

Eg:

Enter the size of array

2

Enter the values of array 1

1      2

3      4



Enter the values of array 2

5      6

7      8

Output:

Sum of array 1 and array 2:

6      8

10     12

```
let arr1 = [];  
let arr2 = [];  
let arr3 = [];  
  
let size1 = parseInt(prompt("Enter the number of rows:"));  
let size2 = parseInt(prompt("enter the coloums  of the array :"));  
  
for (i = 0; i < size1; i++) {  
    arr1[i] = [];  
    for (j = 0; j < size2; j++) {  
        arr1[i][j] = parseInt(prompt("enter the values of the 1st  
array"))  
    }  
}  
  
for (i = 0; i < size1; i++) {  
    arr2[i] = [];  
    for (j = 0; j < size2; j++) {  
        arr2[i][j] = parseInt(prompt("enter the values of the 2nd  
array"))  
    }  
}  
  
for (i = 0; i < size1; i++) {  
    arr3[i] = [];
```

```
for (j = 0; j < size2; j++) {  
    arr3[i][j] = arr1[i][j] + arr2[i][j];  
}  
}  
console.log(arr3);
```

23. Write an object oriented program to store and display the values of a 2D array

a. Program should contains 3 functions including the main function

**main()**

1. Declare an array
2. Call function `getArray()`
3. Call function `displayArray()`

**getArray()**

1. Get values to the array

**displayArray()**

1. Display the array values

Eg:

Enter the size of array

3

Enter the array values

1      2      3

4      5      6

7      8      9

Array elements are:

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |

```
class sample {
    constructor(rows, columns) {
        this.row = rows;
        this.columns = columns;
        this.displayArray = this.getArray();
    }

    getArray() {
        let arr = [];
        for (let i = 0; i < this.row; i++) {
            let arrs = [];
            for (let j = 0; j < this.column; j++) {
                let value = parseInt(prompt("enter the values of
the array"));
                arrs.push(value);
            }
            arr.push(arrs);
        }
        return arr;
    }

    displayArrays() {
        for (let i = 0; i < this.displayArray.length; i++) {
            console.log(this.displayArray[i]);
        }
    }
}

function main() {
```

```
let row = parseInt(prompt("Enter how many rows you want?"));
let columns = parseInt(prompt("how many columns you want :"));
let obj = new sample(row, columns);
obj.displayArrays();
}
main()
```

24. Write a menu driven program to calculate the area of a given object.

- a. Program should contain two classes
  - i. Class 1: MyClass
  - ii. Class 2: Area
- b. Class MyClass should inherit class Area and should contain the following functions
  - i. main()
  - ii. circle()
  - iii. square()
  - iv. rectangle()
  - v. triangle()
- c. Class Area should contain the following functions to calculate the area of different objects
  - i. circle()
  - ii. square()
  - iii. rectangle()
  - iv. triangle()

Class MyClass extends Area{

```
    public static void main(string args[]){

    }
```

```
        circle() {  
  
        }  
  
        square() {  
  
        }  
  
        rectangle() {  
  
        }  
  
        triangle() {  
  
        }  
    }  
}
```

```
Class Area{  
  
    circle(){  
  
    }  
  
    square(){  
  
    }  
  
    rectangle() {  
  
    }  
  
    triangle() {  
  
    }  
}
```

Eg 1:

Enter your choice

1. Circle
2. Square
3. Rectangle
4. Triangle

2

Enter the length

2

Output

Area of the square is: 4

Eg 2:

Enter your choice

1. Circle
2. Square
3. Rectangle
4. Triangle

1

Enter the radius

3

Output

Area of the circle is: 28.26

```
class Area {

    circle(radius) {
        return Math.PI * radius * radius;
    }

    square(side) {
        return side * side;
    }
    rectangle(length, width) {
        return length * width;
    }
    triangle(base, height) {
        return 0.5 * base * height;
    }
}

class MyArea extends Area {
    main() {
        let choice;
        do {
            choice = parseInt(prompt(`
                1. Area of circle \n
                2. Area of square \n
                3. Area of rectangle \n
                4. Area of triangle
                5. Exit
            `))
            switch (choice) {
                case 1: {
                    this.circle()
                    break;
                }
            }
        } while (choice < 5);
    }
}
```

```

        }
        case 2: {
            this.square()
            break;
        }
        case 3: {
            this.rectangle();
            break;
        }
        case 4: {
            this.triangle();
            break;
        }
        case 5: {
            alert('You are Always Welcome')
            break;
        }
        default: {
            alert("Invalid Entry");
        }
    }
} while (choice < 5);
}

circle() {
    let radius = parseInt(prompt("enter radius of the
circle"));
    if (radius <= 0) {
        alert('invalid entry..!')
        return;
    }
    let Result = super.circle(radius)
    console.log(`Area of circle :${Result.toFixed(2)}`)
}

square() {
    let side = parseInt(prompt('enter a side of the Square'));
    if (side <= 0) {
        alert('invalid Entry..!')
        return;
    }
}

```



```

    }
    let result = super.square(side);
    console.log(`Area of the square :${result.toFixed(2)}\`)
  }
  rectangle() {
    let length = parseInt(prompt('enter the length of the
rectangle :'));
    let width = parseInt(prompt('enter the width of the
rectangle :'));
    if (length <= 0 || width <= 0) {
      alert('invalid entry');
      return;
    }
    let result = super.rectangle(length, width);
    console.log(`Area of rectangle :${result.toFixed(2)}\`);
  }
  triangle() {
    let base = parseInt(prompt('enter the base of the triangle
:'));
    let height = parseInt(prompt('enter the height of the
triangle :'));
    if (height <= 0 || base <= 0) {
      alert('Invalid entry');
      return;
    }
    let result = super.triangle(base, height);
    console.log(`Area of the triangle :${result.toFixed(2)}\`);
  }
}

let obj = new MyArea();

obj.main();

```

25. Write a Javascript program to display the status (I.e. display book name, author name & reading status) of books. You are given an object library in the code's template. It contains a list of books with the above mentioned properties. Your task is to display the following:

- If the book is unread:

You still need to read '<book\_name>' by <author\_name>.

- If the book is read:

Already read '<book\_name>' by <author\_name>.

```
var library = [
```

```
{
```

```
  title: 'Bill Gates',
```

```
  author: 'The Road Ahead',
```

```
  readingStatus: true
```

```
},
```

```
{
```

```
  title: 'Steve Jobs',
```

```
  author: 'Walter Isaacson',
```

```
  readingStatus: true
```

```
},
```

```
{
```

```
  title: 'Mockingjay: The Final Book of The Hunger Games',
```

```
author: 'Suzanne Collins',
```

```
readingStatus: false
```

```
}
```

```
];
```

```
var library = [  
  {  
    title: 'Bill Gates',  
    author: 'The Road Ahead',  
    readingStatus: true  
  },  
  {  
    title: 'Steve Jobs',  
    author: 'Walter Isaacson',  
    readingStatus: true  
  },  
  {  
    title: 'Mockingjay: The Final Book of The Hunger Games',  
    author: 'Suzanne Collins',  
    readingStatus: false  
  }  
];  
  
for (i=0; i<library.length; i++) {  
  if (library[i].readingStatus) {  
    console.log(`Already read '${library[i].title}' by  
${library[i].author}.`)  
  } else {  
    console.log(`You still need to read '${library[i].title}'  
by ${library[i].author}.`)  
  }  
}
```

26. Given a variable named `my_string`, *try* reversing the string using `my_string.split().reverse().join()` and then print the reversed string to the console. If the *try* clause has an error, print the error message to the console. Finally, print the *typeof* of the `my_string` variable to the console.

**Output format:**

The statement to print in the *try* block is:

***Reversed string is : \${my\_string}***

The statement to print in the *catch* block is:

***Error : \${err.message}***

The statement to print in the *finally* block is:

***Type of my\_string is : \${typeof my\_string}***

Eg:

**a) Sample Input 0**

"1234"

**Sample Output 0**

Reversed string is : 4321

Type of my\_string is : string

**b) Sample Input 1**

Number(1234)

**Sample Output 1**

Error : my\_string.split is not a function

Type of my\_string is : number

```
try{
  var string = "1234";
  var reversedString = string.split('').reverse().join('');
  console.log(reversedString);
}catch(e) {
  console.log(e)
}
finally{

  console.log('type of my string is :',typeof(string));
}
```

27. Given a variable named `my_height`, you must throw errors under the following conditions:

- `notANumberError`- When `my_height` is NaN
- `HugeHeightError` – When `my_height` is greater than
- `TinyHeightError` - When `my_height` is less than

Eg:

**a) Sample Input 0**

seven

**Sample Output 0**

`notANumberError`

**b) Sample Input 1**

77

**Sample Output 1**

`hugeHeightError`

**c) Sample Input 2**

0

**Sample Output 2**

`tinyHeightError`

**d) Sample Input 3**

8

**Sample Output 3**

8

```
let variable = prompt('enter something');

if (isNaN(variable)) {
    throw new TypeError('notANumberError');
} else if (variable >= 77) {
```

```
    throw new Error('hugeHeightError');
  }else if(variable < 7){
    throw new Error('tinyHeightError');
  }else{
    console.log(variable);
  }
}
```

28. Create a constructor function that satisfies the following conditions:

- a. The name of the constructor function should be *Car*.
- b. It should take three parameters: *name*, *mileage* and *max\_speed*.
- c. Store these parameter values in their respective *this* keywords:  
*this.name*, *this.mileage* and *this.max\_speed*.

```
function car(names,milage,maxSpeed) {
  this.name = names;
  this.milage = milage;
  this.maxSpeed = maxSpeed;
}

let myCar = new car('Hyundai Verna','19-21 km/l','210 kmph');

console.log(myCar.name);
console.log(myCar.milage);
console.log(myCar.maxSpeed);
```

29. Write a *myFilter* function that takes 2 parameters: *myArray* and *callback*. Here, *myArray* is an array of numbers and *callback* is a function that takes the elements of

myArray as its parameter and returns a boolean true if the sum of the number is even or false if the sum of the number is odd.

The myFilter function should return the sum of the array.

**a) Sample Input**

12345

**b) Sample Output**

15

```
let name = [1,2,3,4,5];
let sum = 0;
for(i=0;i<name.length;i++){
    sum = sum + name[i]
}
console.log(sum);
```